Ambrogio Maria Bernardelli

Curriculum Vitae et Studiorum

Research interests: mixed-integer and combinatorial optimization, integrality gaps, stochastic optimization for energy systems.

Research Positions

`01/2025` – `present` Research Fellow (*Assegnista di Ricerca*), Department of Mathematics "F. Casorati", University of Pavia, Italy. Project: *Stochastic optimization for electrical energy storage systems and renewable energy sources*, in collaboration with Eni Plenitude (stochastic optimization models for bidding in intraday electricity markets).

`09/2026` – `12/2026` Visiting Postdoctoral Research Associate, Department of Operations Research & Financial Engineering, Princeton University, Princeton, NJ, USA. Research group of Prof. Bartolomeo Stellato.

Education

Ph.D. in Computational Mathematics and Decision Sciences
University of Pavia and Università della Svizzera Italiana

Cycle: XXXVII.
Dates: 01/2022 – 12/2024.
Thesis title: *Methods for Combinatorial Optimization and Their Applications.*
Defended on: February 20, 2025.
Supervisor: Prof. Stefano Gualandi.

M.Sc. in Mathematics [LM-40]
University of Pavia

Dates: 10/2019 – 09/2021.
Thesis title: *Aspetti Algebrici e Combinatori della Teoria degli Operad* (Algebraic and Combinatorial Aspects of Operad Theory).
Defended on: September 23, 2021.
Supervisor: Prof. Alberto Canonaco.
Grade: 110/110 cum laude.

B.Sc. in Mathematics [L-35]
University of Pavia

Dates: 10/2016 – 09/2019.
Thesis title: *Rappresentazioni Lineari di Gruppi Finiti* (Linear Representations of Finite Groups).
Defended on: September 19, 2019.
Supervisor: Prof. Alberto Canonaco.
Grade: 110/110 cum laude.

Conference Participation

Conferences and workshops where my work was presented. Talks given upon invitation, without submission, are marked with †.

`06/2026` IPCO 2026, Padova, Italy. Poster.

09/2025 The HEXAGON Workshop on power grids, Benevento, Italy†.

09/2025 ODS 2025, Milan, Italy.

07/2024 IPCO 2024, Wrocław, Poland. Poster.

07/2024 ISMP 2024 (presented by Simone Milanesi), Montréal, Canada†.

06/2024 EURO 2024 (presented by Simone Milanesi), Copenhagen, Denmark†.

06/2024 The HEXAGON Workshop on power grids, Bergamo, Italy†.

05/2024 ISCO 2024, San Cristóbal de La Laguna, Spain.

10/2023 2023 INFORMS Annual Meeting (presented by Stefano Gualandi), Phoenix, AZ, USA†.

06/2023 LION17, Nice, France. Paper published in the proceedings [P1].

02/2023 7th AIROYoung Workshop, Milano, Italy. Two talks, one presented by Simone Milanesi.

01/2023 The Mathematics of Machine Learning (presented by Simone Milanesi), Pisa, Italy.

11/2022 Matematica per l'Intelligenza Artificiale e il Machine Learning - Giovani ricercatori, Torino, Italy.

09/2022 YAMC 2022, Arenzano, Italy.

07/2022 ICCOPT / MOPTA 2022, Lehigh University, Bethlehem, PA, USA. The work *Scheduling elective surgeries under uncertainty: a multi-objective stochastic approach* won the second place out of 13 submissions at the 14th AIMMS-MOPTA Optimization Modeling Competition.

Publications

Journal Articles

[J3] Sosso, A., **Bernardelli, A. M.**, Gualandi, S. (2026). The Cloven Traveling Salesman: Cycle Covers and the Integrality Gap of Small ATSP Instances. *Optimization Letters*. doi:10.1007/s11590-026-02312-x.

[J2] **Bernardelli, A. M.**, Gualandi, S., Milanesi, S., Lau, H. C., Yorke-Smith, N. (2025). Multiobjective Linear Ensembles for Robust and Sparse Training of Few-Bit Neural Networks. *INFORMS Journal on Computing*, 37(3), 623–643. doi:10.1287/ijoc.2023.0281.

[J1] **Bernardelli, A. M.**, Bonasera, L., Duma, D., Vercesi, E. (2025). Multi-objective stochastic scheduling of inpatient and outpatient surgeries. *Flexible Services and Manufacturing Journal*, 37(4). doi:10.1007/s10696-024-09542-0.

Conference Proceedings

[P1] **Bernardelli, A. M.**, Gualandi, S., Lau, H. C., Milanesi, S. (2023). The BeMi Stardust: A Structured Ensemble of Binarized Neural Networks. In: Sellmann, M., Tierney, K. (eds) *Learning and Intelligent Optimization. LION 2023*. Lecture Notes in Computer Science, vol 14286, pp. 443–458. Springer, Cham. doi:10.1007/978-3-031-44505-7_30.

Preprints

[A4] **Bernardelli, A. M.** (2026). Cryptanalysis of operad-based cryptography. arXiv preprint arXiv:2611.10134. Under review at *Advances in Mathematics of Communications*.

[A3] Sacilotto, A., **Bernardelli, A. M.**, Gualandi, S. (2026). Emission-aware operation of residential PV-battery systems: the role of information structure. arXiv preprint arXiv:2606.26648. Under review at *Sustainable Energy, Grids and Networks*.

[A2] Riccardi, G., **Bernardelli, A. M.**, Gualandi, S. (2026). Theoretical Perspectives on Jabr-Type Convex Relaxations for AC Optimal Power Flow. arXiv preprint arXiv:2604.00664. Under review as an invited review at *European Journal of Operational Research*.

[A1] **Bernardelli, A. M.**, Vercesi, E., Gualandi, S., Mastrolilli, M., Gambardella, L. M. (2024). Lower bounds for the integrality gap of the bi-directed cut formulation of the Steiner Tree Problem. arXiv preprint arXiv:2405.13773. Under review at *Networks*.

Awards

04/2023 Grant awarded by the TAILOR Connectivity Fund for a one-month visiting period at TU Delft, Delft, The Netherlands.

07/2022 2nd place out of 13 submissions at the 14th AIMMS-MOPTA Optimization Modeling Competition.

09/2019 Department scholarship for the Master's degree program in Mathematics.

Visiting Periods

09/2023 – 02/2024 Università della Svizzera italiana (USI), Lugano, Switzerland. Six-month visit within the joint Ph.D. program, working on integrality gaps with Prof. Luca Maria Gambardella.

04/2023 TU Delft, Delft, The Netherlands. One-month visit at the STAR Lab, working on AI and optimization with Dr. Neil Yorke-Smith.

Conference Organization

05/2025 Co-organizer of the COMPMAT Spring Workshop 2025, University of Pavia, May 23, 2025.

02/2025 Member of the organizing committee of the 9th AIROYoung Workshop, University of Pavia, February 26–28, 2025.

2022--2024 Co-organizer of Caffè Beltrami, a seminar series introducing different areas of research in mathematics to Bachelor's and Master's students.

Teaching Activities

Teaching

2026/2027 511180 – Algorithms for Optimization (M.Sc. in Finance, LM-16), University of Pavia, 22 hours. Co-teaching with Prof. Davide Duma, who held the other 22 hours of the course.

2024/2025 511180 – Algorithms for Optimization (M.Sc. in Finance, LM-16), University of Pavia, 22 hours. Co-teaching with Prof. Daniele Boffi, who held the other 22 hours of the course.

Supervising

2025/2026 *Ottimizzazione Stocastica Multistadio per la Gestione del Rischio nel Mercato Energetico Italiano*, Silvia Maria Teresa Stimolo, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2024/2025 *Riduzione delle Emissioni di Carbonio mediante Ottimizzazione Stocastica dello Stoccaggio Energetico a Batteria in Sistemi Fotovoltaici*, Anna Sacilotto, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2024/2025 *Strengthening the SOCP formulation for the ACOPF problem*, Arthur Mazeyrat, Master's Report in Operations Research and Combinatorial Optimization at University Grenoble Alpes, co-supervised with Prof. Stefano Gualandi and Gabor Riccardi.

2023/2024 *The Cloven Travelling Salesman: a new Approach to the ATSP Integrality Gap Estimation*, Alessandro Sosso, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2022/2023 *On the exactness of Jabr-like models and distributionally robust stochastic optimal power flow*, Gabor Riccardi, Master's Thesis in Mathematics at University of Pavia, co-supervised with Prof. Stefano Gualandi.

2022/2023 *Analisi di architetture per Binarized Neural Networks*, Andrea Panno, Bachelor's Thesis in Bioengineering at University of Pavia, co-supervised with Prof. Stefano Gualandi and Simone Milanesi.

Tutoring

2021/2022 Tutoring activity for the course *Geometria e Algebra* that was held by prof. Francesco Bonsante for the Bachelor's degree program *Bioingegneria* at University of Pavia. 40 hours.

2021/2022 Tutoring activity for first-year students attending the Bachelor's degree program in Mathematics at the University of Pavia. 40 hours.

Personal Information

Year of birth: 1997

Email: ambrogiomaria.bernardelli(at)unipv.it

Personal webpage: ambrogiomb.github.io

ORCID: 0000-0002-2328-7062

Research group website: compopt.it

Last updated: November 10, 2026